

MA6.301 (Monsoon 2026)

Probability and Statistics

Assignment 2

Distributions, Expectation, and Joint Random Variables

Instructor: Naresh Manwani
Deadline: TBA
Total Marks: 100 (10% of the course)

Instructions

1. **All questions are compulsory.** Show all relevant working and clearly justify each step. State the axiom, definition, theorem, or result being used wherever applicable.
2. The assignment must be submitted in **both hard copy and soft copy**. The hard copy must be entirely **handwritten**; typed or printed submissions will not be accepted. The soft copy must be uploaded to Moodle as a single PDF named `rollnumber_a2.pdf`.
3. **Group discussion and consultation of reference materials are encouraged** for developing understanding and solving problems. However, the final solution and write-up must be your own **individual work**.
4. If you substantially use any textbook, lecture notes, website, paper, or other reference material while preparing your solution, you must **properly cite/reference the source** in your submission.
5. **Plagiarism or copying will not be tolerated.** Any submission found to contain plagiarised or copied work may receive **zero marks for the assignment/component** and, in serious or repeated cases, the penalty may be extended to an **F grade in the course**, as per course/institute policy.
6. State the support of every PMF or PDF you obtain. Whenever a CDF is requested, give it for **all real arguments**, including values outside the support.
7. Throughout this assignment:
 - Geometric(p) counts trials **up to and including the first success**.
 - Exp(λ) uses the **rate** parameter.
 - In $N(\mu, \sigma^2)$, the second parameter is the **variance**.
 - $G_X(t) = \mathbb{E}[t^X]$ denotes a probability generating function.
 - $\Phi(z)$ denotes the standard normal CDF.
8. You may use the fact that a PGF uniquely determines the distribution of a nonnegative integer-valued random variable.

Suggested references

- Hossein Pishro-Nik, *Introduction to Probability, Statistics, and Random Processes*, available at <https://www.probabilitycourse.com/>: sections on random variables, expectation, conditional distributions, and joint distributions.
- Dimitri P. Bertsekas and John N. Tsitsiklis, *Introduction to Probability*, 2nd ed., Athena Scientific, 2008: chapters on discrete and general random variables.

The questions below are original exercises aligned with the supplied syllabus, rather than reproductions of exercises from these references.

Question 1. [12 marks] Geometric waiting time with a retry limit

A device makes successive transmission attempts. Each attempt succeeds with probability $p \in (0, 1)$, independently of all other attempts.

Let X be the number of attempts required for the first success under an unlimited-attempt policy. A practical policy allows at most m attempts, where $m \geq 2$ is an integer. Thus, the actual number of attempts made under this policy is

$$T = \min(X, m).$$

The transmission succeeds under the practical policy exactly when $X \leq m$.

(a) [2] Starting from the geometric PMF or tail probability, prove that

$$P(X > r + s \mid X > r) = P(X > s)$$

for all nonnegative integers r, s . Give a one-sentence interpretation in the transmission setting.

(b) [4] Using the partition $\{X = 1\}, \{X > 1\}$, total expectation, and memorylessness, derive

$$\mathbb{E}[X], \quad \mathbb{E}[X^2], \quad \text{Var}(X).$$

You may assume these moments are finite. Do not quote their known formulas or evaluate the corresponding infinite moment sums directly.

(c) [3] Find the PMF and CDF of T . Pay particular attention to the probability at $T = m$.

You may use $\lfloor t \rfloor$, the greatest integer not exceeding t , when writing the CDF.

(d) [3] Use memorylessness and total expectation to obtain a closed-form expression for $\mathbb{E}[T]$. Also find

$$P(X \leq m \mid T = m).$$

Explain briefly why making exactly m attempts does not necessarily mean that the transmission failed.

Question 2. [10 marks] Combining batches and approximating rare defects

A manufacturer produces two batches containing 800 and 1200 components, respectively. Every component is defective with probability 0.001, independently of all other components.

Let D_1 and D_2 be the numbers of defective components in the two batches, and let

$$D = D_1 + D_2.$$

(a) [3] Starting from the binomial PMF, derive $G_{D_1}(t)$ and $G_{D_2}(t)$. Use independence and the product rule for PGFs to obtain $G_D(t)$, and identify the exact distribution of D .

(b) [3] Find an exact expression for

$$P(D \geq 2 \mid D \geq 1).$$

Your final expression should not contain an unevaluated sum.

(c) [4] Approximate the probability in part (b) using a Poisson distribution.

- State the Poisson parameter and explain why this approximation is appropriate.
- Evaluate both the exact probability and the Poisson approximation to **six decimal places**.
- Report the absolute approximation error

$$|\text{Poisson approximation} - \text{exact probability}|$$

to six decimal places.

Question 3. [12 marks] Independent Poisson counts conditioned on their total

During a fixed observation period, three independent sources generate event counts

$$X_1 \sim \text{Poisson}(2), \quad X_2 \sim \text{Poisson}(3), \quad X_3 \sim \text{Poisson}(1).$$

Let

$$N = X_1 + X_2 + X_3.$$

(a) [3] Derive the PGF of a $\text{Poisson}(\lambda)$ random variable directly from its PMF. Then use PGFs to identify the distribution of N .

(b) [4] For an integer $n \geq 1$, derive

$$P(X_1 = x_1, X_2 = x_2, X_3 = x_3 \mid N = n).$$

State all conditions on x_1, x_2, x_3 , including when this conditional probability is zero. Identify the resulting multinomial distribution and its parameters.

(c) [2] Use part (b) to identify the conditional distribution of X_1 given $N = 6$. Hence find

$$P(X_1 = 2 \mid N = 6) \quad \text{and} \quad \mathbb{E}[X_1 \mid N = 6].$$

(d) [3] Are X_1 and X_2 conditionally independent given $N = 6$?

Justify your answer by choosing specific values a, b and comparing

$$P(X_1 = a, X_2 = b \mid N = 6)$$

with

$$P(X_1 = a \mid N = 6)P(X_2 = b \mid N = 6).$$

Merely stating that “the total is fixed” is not sufficient.

Question 4. [8 marks] A sum of binomials with different success probabilities

Let

$$X \sim \text{Binomial}\left(2, \frac{1}{2}\right), \quad Y \sim \text{Binomial}\left(2, \frac{1}{4}\right),$$

and suppose X and Y are independent. Define

$$S = X + Y.$$

(a) [3] Use discrete convolution to obtain the complete PMF of S . Show the convolution formula before evaluating the probabilities.

(b) [2] Calculate $\mathbb{E}[S]$ and $\text{Var}(S)$, using the means and variances of X and Y .

(c) [3] Determine whether there exists $r \in [0, 1]$ such that

$$S \sim \text{Binomial}(4, r).$$

Give a proof using the mean and variance from part (b). Your answer should explain why independence alone does not guarantee that a sum of binomial random variables is binomial.

Question 5. [10 marks] Dependence that is not detected by a product of expectations

The joint PMF of X and Y is

$$p_{X,Y}(x, y) = \begin{cases} c(x^2 + y^2), & x, y \in \{-1, 0, 1\}, \\ 0, & \text{otherwise,} \end{cases}$$

where c is a positive constant.

- (a) [2] Find c , and obtain the marginal PMFs of X and Y .
- (b) [3] Find the conditional PMF of X given $Y = 0$. Use it to determine whether X and Y are independent. Support your conclusion with an explicit probability comparison.
- (c) [3] Calculate

$$\mathbb{E}[X], \quad \mathbb{E}[Y], \quad \mathbb{E}[XY].$$

Use your results and part (b) to determine whether

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$$

is sufficient to establish independence.

- (d) [2] Calculate $\text{Var}(X + Y)$ using

$$\text{Var}(X + Y) = \mathbb{E}[(X + Y)^2] - (\mathbb{E}[X + Y])^2.$$

Compare your answer with $\text{Var}(X) + \text{Var}(Y)$. Does equality of these quantities establish independence in this example?

Question 6. [10 marks] Empty servers and indicator variables

There are $m \geq 1$ distinguishable jobs and $n \geq 2$ distinguishable servers. Each job is assigned independently and uniformly to one of the n servers.

For $i = 1, \dots, n$, define

$$I_i = \begin{cases} 1, & \text{server } i \text{ receives no jobs,} \\ 0, & \text{otherwise.} \end{cases}$$

Let

$$K = \sum_{i=1}^n I_i$$

be the number of empty servers. Let L be the number of servers receiving exactly one job.

- (a) [3] Use indicator variables to derive closed-form expressions for $\mathbb{E}[K]$ and $\mathbb{E}[L]$. Define any additional indicators you introduce.
- (b) [2] For distinct servers $i \neq j$, find

$$P(I_i = 1, I_j = 1).$$

Determine whether I_i and I_j are independent, and justify your conclusion algebraically.

- (c) [4] Derive a closed-form expression for $\text{Var}(K)$. Do not assume the indicators are independent.

Hint: Expand K^2 , use $I_i^2 = I_i$, and observe that

$$\mathbb{E}[I_i I_j] = P(I_i = 1, I_j = 1).$$

- (d) [1] Now let $m = n$. Find

$$\lim_{n \rightarrow \infty} \mathbb{E}\left[\frac{n - K}{n}\right],$$

the limiting expected fraction of servers receiving at least one job.

Question 7. [10 marks] Functions of a uniform random variable

Let

$$U \sim \text{Uniform}[-2, 4].$$

- (a) [2] Write the CDF $F_U(u)$, and calculate

$$P(-1 < U \leq 2 \mid U > 0).$$

(b) [3] Define $Y = U^2$. Use integration with respect to the density of U to find

$$\mathbb{E}[Y] \quad \text{and} \quad \text{Var}(Y).$$

You are not required to find the density of Y .

(c) [3] Define

$$W = \begin{cases} -1, & U < 0, \\ 0, & 0 \leq U \leq 1, \\ 1, & U > 1. \end{cases}$$

Find the complete PMF of W , its mean, and its variance.

(d) [2] Find the conditional density $f_{U|U>0}(u)$, and use it to calculate

$$\mathbb{E}[U \mid U > 0].$$

Question 8. [8 marks] Normal measurements and a decreasing calibration rule

The temperature X , measured in degrees Celsius, satisfies

$$X \sim N(20, 4).$$

A sensor converts the numerical temperature reading into a dimensionless index

$$Y = 100 - 2X.$$

(a) [2] State the distribution of Y , including its mean, variance, and standard deviation.

(b) [3] Calculate

$$P(Y \geq 64 \mid X \leq 22).$$

Express the answer in terms of Φ before giving its numerical value. Clearly show how the condition $Y \geq 64$ translates into a condition on X .

(c) [3] Find the unique $d \geq 0$ satisfying

$$P(|Y - \mathbb{E}[Y]| \leq d) = 0.95.$$

Give d to three decimal places, and state the corresponding central (95%) interval for Y .

Question 9. [8 marks] Exponential lifetime and a capped operating period

The lifetime X , measured in hours, of a component satisfies

$$X \sim \text{Exp}\left(\frac{1}{5}\right).$$

(a) [3] Derive the survival probability $P(X > x)$ for $x \geq 0$ from the exponential density. Then prove that

$$P(X > s + t \mid X > s) = P(X > t)$$

for all $s, t \geq 0$.

(b) [2] Given that the component has survived for more than 3 hours, find its conditional density and calculate

$$\mathbb{E}[X \mid X > 3] \quad \text{and} \quad \mathbb{E}[X - 3 \mid X > 3].$$

State which expectation represents the remaining lifetime.

(c) [3] The component is removed from service at failure or at 8 hours, whichever occurs first. Its operating time is therefore

$$T = \min(X, 8).$$

Find the CDF of T and calculate $P(T = 8)$. Does there exist an ordinary PDF f_T satisfying

$$P(a < T \leq b) = \int_a^b f_T(t) dt$$

for every $a < b$? Justify your answer using the probability at $T = 8$.

Question 10. [12 marks] A joint density on a triangular region

The joint PDF of X and Y is

$$f_{X,Y}(x,y) = \begin{cases} c, & 0 < x < y < 2, \\ 0, & \text{otherwise,} \end{cases}$$

where c is a positive constant.

- (a) [2] Sketch the support and determine c by normalization.
- (b) [3] Find the marginal PDFs $f_X(x)$ and $f_Y(y)$. Determine whether X and Y are independent. For your independence argument, compare the probability of one explicitly chosen joint event with the product of its marginal event probabilities.

- (c) [3] Calculate

$$P(Y \geq 2X, X + Y \leq 2).$$

Sketch the relevant region and write the integration limits explicitly before evaluating the integral.

- (d) [4] Let $A = \{Y \leq 1\}$. Find

$$P(A), \quad f_{X|A}(x), \quad \mathbb{E}[X | A].$$

Verify that your conditional density integrates to one.

Hint: To obtain $f_{X|A}$, integrate the joint density over the values of y compatible with both the original support and A , then normalize by $P(A)$.